C 23887 **(Pages : 2)** **Name.....**

Reg. No.....

SECOND SEMESTER (CBCSS–UG) DEGREE EXAMINATION, APRIL 2022

Physics/Applied Physics

PHY 2B 02/APH 2B 02—MECHANICS

(2019 Admissions)

Time : Two Hours Maximum : 60 Marks

The symbols used in this question paper have their usual meanings.

Section A (Short Answer Type)

Answer **all** *questions in two or three sentences.*
Each correct answer carries a maximum of 2 *marks.*

1. Briefly explain the characteristic impedance of a travelling wave.
2. State and explain the principle of equivalence.
3. Discuss the condition for nondispersive wave.
4. Write down the general expression for a plane progressive wave traveling in : (a) Positive x direction ; and (b) Negative x direction.
5. Write two important properties of travelling waves.
6. Define the time average of a function $f(t)$ with proper diagrams.
7. With examples explain simple harmonic motion
8. Define fictious force. Deduce an expression for the force experienced by a particle in a co-ordinate system with uniform acceleration A.
9. Explain Coriolis force. What is the effect of Coriolis force on wind moving over the surface of earth ?
10. Define elasticity.
11. Write down the equation for phase velocity in terms of : a) Frequency ; and b) Wave length.
12. Define fictious force ? Deduce an expression for the force experienced by a particle in a co-ordinate system with uniform acceleration A.

(Ceiling-20)

Turn over

Section B (Paragraph / Problem Type)

Answer **all** *questions in a paragraph of about half a page to one page.*
Each correct answer carries a maximum of 5 *marks.*

13. Write note on : a) Exponential wave ; and b) Exponential zig zag wave.

14. With figure explain transmission line low pass filter. What is its phase velocity ?

15. A small weight of mass *m* hangs from a string in an automobile which accelerates at rate A. What is the static angle of the string from the vertical, and what is its tension ? Analyse the problem both in an inertial frame and in a frame accelerating with the car.

16. Define Q factor of an oscillator. A musician's tuning fork rings at A above middle C, 440 Hz. A sound level meter indicates that the sound intensity decreases by a factor of 5 in 4 s. Calculate the Q of the tuning fork.

17. The centre of mass of a 1600 kg car is midway between the wheels and 0.7 m above the ground. The wheels are 2.6 m apart :
    (a) What is the minimum acceleration A of the car so that the front wheels just begin to lift off the ground ?
    (b) If the car decelerates at rate g, what is the normal force on the front wheels and on the rear wheels?

18. A satellite of mass *m* = 2000kg is in elliptic orbit about the earth. At perigee it has an altitude of 1100km and at apogee its altitude is 4100km. Calculate the energy needed to put the satellite into orbit by neglecting losses due to friction.

19. Define Q factor of an oscillator. A musician's tuning fork rings at A above middle C, 440 Hz. A sound level meter indicates that the sound intensity decreases by a factor of 5 in 4 s. Calculate the Q of the tuning fork.

(Ceiling - 30)

Section C (Essay Type)

Essays.
Answer in about two pages, any one question.
Answer carries 10 *marks.*

20. Explain and derive equation for velocity of sound wave in air. What is Newton's correction ?

21. State Fourier's theorem. Determine the values of Fourier's co-efficients. What are conditions of its applicability ? Discuss Fourier analysis of a non-periodic function with suitable plots.

(1 × 10 = 10 marks)